# The Mathematics of the Thesis

## Identifiability, Smile Transport and Finite-Sample Model-Risk Certification for the JIBAR-to-ZARONIA Derivatives Book

Revised after council review, and re-cut to a three-year programme

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9 September 2026

**Abstract**

This document states every mathematical result the thesis promises, at proposal precision: hypotheses, statement, proof route, failure mode. It is the third version. The first version was put before three reviewers acting as a mathematical statistician, a mathematical finance theorist and a doctoral examiner; their corrections are incorporated and marked in the margin. Two corrections were structural. First, the direction of the identifiability argument in the finance half was reversed: the unidentified correlations contaminate the observed JIBAR smile, not the target compounded-rate smile. Second, the statistics half as first drafted assembled known results and would not carry a mathematical-statistics doctorate; a new theorem on the calibration-conditional coverage error over a mixing class, with the optimal block length of the blocked route as its first part (Theorem 9), is now the spine. The third version corrects Theorem 9 after a literature scan: the blocked rate $N^{-r/(2r+3)}$ of the second version is not minimax in any coverage formulation (Barber and Pananjady 2026; part (ii) below), and the theorem is restated for the calibration-conditional coverage error at confidence $\delta$, where mixing is provably the binding constraint. The final section re-cuts the programme to thirty-six months.

## Summary of council corrections

| Result | Change forced by review |
|---|---|
| Lemma 1 | Same-$g$ scaling is valid without Dambis–Dubins–Schwarz; linear $g$ and the factor $\Delta/3$ are assumptions, not theorems. Non-decaying vol-of-vol added as Proposition 2; its claimed 17% curvature understatement was wrong (total variance is not the smile) and is corrected to a 0.002 bp effect. |
| Theorem 3 | JIBAR caplet is priced under $\mathbb{Q}^{T+\Delta}$, not $\mathbb{Q}^{T}$; the in-advance/in-arrears convexity term does not exist and is deleted. Basis correlations enter the JIBAR smile, not the compounded caplet. Two unidentified correlations, not one. Expansion parameter redefined. |
| Theorem 4 | Becomes a corollary of Theorem 3; difference is at the money, not wing-only; rank condition is trivial. Demoted from difficulty 4 to 2–3. |
| Proposition 5 | "Static arbitrage" replaced by "model inconsistency"; exact only for normal SABR with equal volatilities; otherwise leading order in variance. Demoted to proposition, appendix in the three-year plan. |
| Theorem 6 | Coupling cost $(n+1)\beta(b)$, not $2n\beta(b)$; deployment block must be gap-separated; conditional form uses the Beta law of conformal coverage, not DKW; drift penalty is Kolmogorov distance $d_{\mathrm{K}} \le d_{\mathrm{TV}}$, applied after coupling. |
| Proposition 7 | The first version was false (counterexample: point mass versus narrow uniform). Correct bound is $\sqrt{2LW_1}$; plug-in rate $n^{-1/4}$; the dependent-data version of the empirical $W_1$ bound is currently missing. |
| Theorem 8 | $\delta_{+\infty}$ is not a measure on $\mathbb{R}$; construction replaced; the sharp constant is 1, not $1-\alpha$; the argument is adversarial, not Le Cam. |
| First-draft bootstrap theorem | Dropped as a theorem. Empirical comparison with block bootstrap retained. |
| Theorem 9 | **New in the second version; corrected in the third.** The claimed minimax rate $N^{-r/(2r+3)}$ was false: Barber and Pananjady (2026) obtain the marginal deficit $N^{-r/(r+1)}$ over $\mathcal{B}_r$ without blocking, and the unblocked sample quantile has conditional coverage error $N^{-1/2}$ for every $r > 1$. The deficit $\sqrt{b/N} + Nb^{-(r+1)}$ comes from Theorem 6(ii), not (i). Exact minimiser $(2(r+1))^{2/(2r+3)}N^{3/(2r+3)}$ added. Restated as the minimax calibration-conditional coverage error at confidence $\delta$ over $\mathcal{B}_r$, with a mixing-driven lower bound; positioned against Halkiewicz (2026), whose lower bound is drift-driven. This is the doctoral theorem. |

# 1 Setting and notation

Fix a filtered probability space carrying a Brownian motion $(W, Z)$ with $d\langle W, Z\rangle_t = \rho\, dt$, and a third Brownian motion $W^B$ with $d\langle W^B, W\rangle_t = \rho_{BF}\, dt$, $d\langle W^B, Z\rangle_t = \rho_{B\alpha}\, dt$. Let $P(t, U)$ denote the zero-coupon bond price and $\mathbb{Q}^U$ the $U$-forward measure.

**Compounded rate.** For an accrual window $[T, T + \Delta]$ with daily fixings $r_i$ and day-count fractions $\delta_i$,

$$R(T, T+\Delta) = \frac{1}{\Delta}\Big(\prod_i (1 + r_i\delta_i) - 1\Big) \;\approx\; \frac{1}{\Delta}\Big(\exp\Big(\int_T^{T+\Delta} r_u\, du\Big) - 1\Big). \tag{1}$$

The continuous approximation is used throughout; it makes $1+\Delta F_t = P(t,T)/P(t,T+\Delta)$ exact and differs from the daily product at second order in $r_i\delta_i$, below a basis point. [Council correction: state once, keep continuous form]

**Forwards.** The forward compounded rate $F_t := \mathbb{E}^{\mathbb{Q}^{T+\Delta}}[R(T,T+\Delta) \mid \mathcal{F}_t]$ is a $\mathbb{Q}^{T+\Delta}$-martingale. The JIBAR forward $L_t$ for the same window is fixed at $T$ and *paid at* $T+\Delta$; it is therefore also a $\mathbb{Q}^{T+\Delta}$-martingale, and so is the basis $B_t := L_t - F_t - s$, where $s$ is the fixed credit adjustment spread. In particular $\mathbb{E}^{\mathbb{Q}^{T+\Delta}}[B_T] = B_0 = L_0 - F_0 - s$ is read off the two curves. [Council correction: both caplets live under $\mathbb{Q}^{T+\Delta}$; no in-advance/in-arrears convexity term]

**Decay and clock.** Let $g$ be deterministic and bounded with $g(t) = 1$ for $t \le T$ and $g(t) \downarrow 0$ on $(T, T+\Delta]$, and set $\tau(t) := \int_0^t g(u)^2\,du$.

**Assumption 1** (Linear decay)**.** $g(t) = (T+\Delta-t)/\Delta$ on $(T, T+\Delta]$, so that $\tau(T+\Delta) = T+\Delta/3$.

[Council correction: linear $g$ is exact only when the instantaneous forward-rate volatility is flat in maturity; generically $\tau(T+\Delta) = T + c_g\Delta$ with $c_g \in (0,1)$. Every result below inherits this assumption.]

**Dynamics.** Under $\mathbb{Q}^{T+\Delta}$,

$$dF_t = g(t)\,\alpha_t F_t^\beta\,dW_t, \qquad d\alpha_t = g(t)\,\nu\,\alpha_t\,dZ_t, \qquad dB_t = \eta\,dW_t^B. \tag{2}$$

The dimensionless basis scale is $\varepsilon := \eta/(\alpha_0 F_0^\beta)$. [Council correction: $\eta\sqrt{T}$ is a rate, not a small parameter]

**Scores.** For a fixed valuation-and-hedging policy $\pi$, the hedging error over a block of $b$ consecutive rebalancing periods is the *score* $S$. Calibration scores have law $P$, deployment scores law $Q$. $F_P, F_Q$ are their distribution functions, $d_{\mathrm{K}}(P,Q) = \sup_x |F_P(x) - F_Q(x)| \le d_{\mathrm{TV}}(P,Q)$ the Kolmogorov distance.

## 2 Finance results

**Lemma 1** (Time-changed SABR; Willems 2020, Lyashenko–Mercurio 2019)**.** *Under* (2) *with the same $g$ on both diffusion coefficients, $(F_t, \alpha_t) = (\tilde{F}_{\tau(t)}, \tilde{\alpha}_{\tau(t)})$ where $(\tilde{F}, \tilde{\alpha})$ is constant-parameter SABR$(\alpha_0, \beta, \rho, \nu)$. Hence*

$$P(0,T+\Delta)\,\mathbb{E}^{\mathbb{Q}^{T+\Delta}}\big[(F_{T+\Delta} - K')^+\big] = P(0,T+\Delta)\,C^{\mathrm{SABR}}\big(F_0, K', \tau(T+\Delta); \alpha_0, \beta, \rho, \nu\big), \tag{3}$$

*with the Hagan approximation carrying its usual $O(\alpha^2\tau)$ error.*

*Proof.* Because $\tau$ is deterministic, $(\int_0^t g\,dW, \int_0^t g\,dZ)$ is a correlated Brownian pair in the clock $\tau$; substitute. [Council correction: no Dambis–Dubins–Schwarz needed; the same-$g$ condition is what makes the substitution exact] □

**Proposition 2** (Non-decaying vol-of-vol; target)**.** *If instead $d\alpha_t = \nu\alpha_t\,dZ_t$ (vol-of-vol does not run off, the physically defensible choice), then in the clock $\tau$ the vol-of-vol is $\nu/g(\tau^{-1}(u))$ and no closed form exists.*

*(i) (Total variance.) The total vol-of-vol variance is $\nu^2(T+\Delta)$ rather than $\nu^2(T+\Delta/3)$: an excess of $\frac{2}{3}\nu^2\Delta$, which is $15.4\%$ of $\nu^2\tau^*$ and $16.7\%$ of $\nu^2 T$ for $T = 1y$, $\Delta = 3m$. This quantity is the variance of $\log\alpha_{T+\Delta}$; it does not by itself determine the smile.*

*(ii) (Smile-relevant effective parameters.) At leading order the Hagan–Lesniewski–Woodward effective parameters, computed in calendar time with $du = g^2\,dt$ and $\int_T^{T+\Delta} g^{2k}\,dt = \Delta/(2k+1)$, satisfy*

$$\frac{\nu_{\mathrm{eff}}^2}{\nu^2} = 1 + \frac{2\Delta^3/567}{\tau^{*3}/3} = 1.00013, \qquad \frac{\rho_{\mathrm{eff}}\nu_{\mathrm{eff}}}{\rho\nu} = 1 + \frac{\Delta^2/90}{\tau^{*2}/2} = 1.0012$$

*for $T = 1y$, $\Delta = 3m$, so the two variants agree to about $0.002$ bp at $\pm 50$ bp from the money. Lemma 1 does not understate smile curvature in any material way.*

[Council correction: the second version claimed a 17% curvature understatement by identifying curvature with total vol-of-vol variance; the forward-measures referee showed the smile-relevant effective parameters differ by $0.013\%$ and $0.12\%$, confirmed numerically by the rates code. Whether the leading-order weights coincide with Hagan et al. (2002), Appendix B, is a librarian task.]

*Remark.* Neither result is a contribution. Lemma 1 is cited; Proposition 2 is a two-page derivation in Chapter 2 and fixes which variant the empirical chapters use.

**Theorem 3** (Identified and unidentified components of the smile map; target)**.** *Let (2) hold with Assumption 1, fix $\beta$, and suppose $\nu \neq 0$. Denote by $\sigma_L^N(K,T)$ the observed JIBAR normal implied volatility and by $\sigma_F^N(K', T+\Delta)$ the compounded-rate normal implied volatility at the fallback strike $K' = K - s$. Then*

$$\sigma_F^N(K', T+\Delta) \;=\; \Phi_0\big(K';\, \sigma_L^N(\cdot,T),\, s,\, B_0,\, \Delta\big) \;-\; \rho_{BF}\,\eta\,\Phi_1(K') \;-\; \rho_{B\alpha}\,\eta\,\nu\,\Phi_2(K') \;+\; O(\varepsilon^2), \quad (4)$$

*where*

1. *$\Phi_0$ is the SABR smile at clock $T+\Delta/3$ whose triple $(\alpha_0, \rho, \nu)$ is fitted to $\sigma_L^N(K'+s+B_0,\, T)$ at clock $T$: a strike shift by $s + B_0$ and the time change of Lemma 1;*
2. *$\Phi_1(K') = 1 + O(K' - F_0)$: a parallel level shift;*
3. *$\Phi_2(K') \propto (K' - F_0)$: a skew tilt.*

*Consequently: the curvature parameter $\nu$ and the map $\Phi_0$ are identified from the observable set $\mathcal{S} = \{\sigma_L^N(\cdot,T)\} \cup \{B_0\} \cup \{s\}$; the pair $(\eta\rho_{BF}, \eta\rho_{B\alpha})$ is not identified at $O(\varepsilon)$, and $(\eta, \rho_{BF}, \rho_{B\alpha})$ is not identified at $O(\varepsilon^2)$. In particular the successor at-the-money volatility is identified only within an interval of half-width $\eta$.*

[Council correction: the first draft had the unidentified correlation entering the compounded caplet; it is the other way round. The compounded caplet depends only on $(\alpha_0, \beta, \rho, \nu)$. The JIBAR smile is a contaminated view of that triple, because $L = F + B + s$ and the basis noise shifts its level (through $\rho_{BF}$) and tilts its skew (through $\rho_{B\alpha}\nu$). The convexity term was deleted because both instruments pay at $T + \Delta$.]

*Proof route.* Under $\mathbb{Q}^{T+\Delta}$ write $L = F + B + s$ and expand the JIBAR caplet in $\varepsilon$: the level of $\sigma_L^N$ is $\sigma_F^N$-at-clock-$T$ plus $\rho_{BF}\eta$ at first order (variance of a sum), the skew acquires $\rho_{B\alpha}\eta\nu$ from $\mathrm{Cov}(B, \alpha)$, and the curvature is unaffected at $O(\varepsilon)$. Invert: fitting SABR to $\sigma_L^N$ returns $(\alpha_0 - \rho_{BF}\eta + \ldots,\; \rho + c\,\rho_{B\alpha}\eta\nu/\alpha_0 + \ldots,\; \nu)$. Push the fitted triple through the time change to obtain $\Phi_0$; the residual terms are $\Phi_1, \Phi_2$.

*Failure mode.* If the basis has mean reversion comparable to $1/T$, the $O(\varepsilon)$ coefficients acquire $T$-dependence and the interval half-width becomes $\eta\, h(\kappa T)$ for an explicit $h$. The statement survives with a modified constant.

**Theorem 4** (Two models; corollary of Theorem 3)**.** *For any $(v_1, v_2)$ with $|v_i| \leq \eta$ there exist models $\mathcal{M}_\pm$ with $(\eta\rho_{BF}, \eta\rho_{B\alpha}) = \pm(v_1, v_2)$ that match every JIBAR caplet to $O(\varepsilon^2)$ and match $B_0$ exactly, while their compounded-rate at-the-money volatilities differ by $2v_1$ and their skews by $2v_2\nu$, at every strike.*

*Proof.* Flip the sign of $(\rho_{BF}, \rho_{B\alpha})$ and re-solve $(\alpha_0, \rho)$ so that the first-order level and skew of $\sigma_L^N$ are unchanged. The Jacobian $\partial(\mathrm{level}, \mathrm{skew})/\partial(\alpha_0, \rho)$ is nonsingular for $\nu \neq 0$, so the re-solve exists. Read the differences off (4). [Council correction: rank is immediate, not a research question; difference is at the money, not wing-only] □

*Remark* (Parameter count). With $m$ expiries and $\beta$ fixed: observables are $3m$ smile features (level, skew, curvature per expiry; additional strikes test SABR shape and carry $\eta$ only at $O(\varepsilon^2)$)

plus $m$ basis points, which pin $B_{0,i}$ and hence strike shifts but no smile parameters. Unknowns are $3m$ SABR parameters plus $(\eta, \rho_{BF}, \rho_{B\alpha})$ if shared across expiries. Deficit: 3 (two at $O(\varepsilon)$). If the basis parameters are per-expiry, the deficit is $3m$. Historical basis data estimate the nuisance under the physical measure, not the pricing measure; that estimate is the ambiguity set that Theorem 6 takes as input.

**Proposition 5** (Tenor consistency; target, appendix in the three-year plan)**.** *For $t \le T_1 < T_2 < T_3$, $(1 + \Delta_{13}F_t^{13}) = (1 + \Delta_{12}F_t^{12})(1 + \Delta_{23}F_t^{23})$ exactly (ratios of zero-coupon bonds).*

*(a) If $\beta = 0$, $g$ linear and $(W, Z)$ common, the induced $R_{13}$ is time-changed normal SABR with linear decay if and only if $\alpha_{12} = \alpha_{23}$; then $\alpha_{13} = \alpha_{12}$, $(\rho, \nu)$ are inherited, and*

$$\Delta_{13}^2\tau_{13} = \Delta_{12}^2\tau_{12} + \Delta_{23}^2\tau_{23} + 2\Delta_{12}\Delta_{23}\, c, \qquad c = T_1 + \Delta_{12}/2,$$

*exactly; the clocks compose.*

*(b) Otherwise the displayed relation is the variance-matched projection, valid to leading order in $\alpha^2\tau$, and the constraints tying $(\rho, \nu)$ across tenors are approximate.*

*(c) Violation of (a)–(b) means no model in the class calibrates jointly to the 3M and 6M smiles. It is not a static arbitrage: $(R_{13} - K)^+$ is not spanned by 3M caplets.* [Council correction: a model-free statement would need Hobson–Laurence–Wang-type bounds for an option on a product with given marginals, which is a different theorem and is not claimed]

## 3 Statistics results

**Assumption 2** (Mixing and separation)**.** The per-period hedging-error process of the fixed policy $\pi$ is $\beta$-mixing with coefficients $\beta(k)$ (non-stationary definition: supremum over time origins) and stationary on the calibration window. The fitting sample $\mathcal{D}_{\text{fit}}$, the $n$ calibration blocks of length $b$, and the deployment block are pairwise separated by gaps of length at least $b$. [Council correction: the deployment gap is required; without it $S_0$ is not controlled at all]

Let $k := \lceil (1-\alpha)(n+1) \rceil$, $\hat{q} := S_{(k)}$ the $k$-th order statistic of the calibration scores, and $b_{n,k}(\delta) := \mathrm{Beta}^{-1}(\delta;\, k,\, n+1-k)$ the $\delta$-quantile of the $\mathrm{Beta}(k, n+1-k)$ law.

**Theorem 6** (Coverage under mixing and drift; target)**.** *Under Assumption 2, with calibration scores $S_1, \dots, S_n \sim P$ marginally and deployment score $S_0 \sim Q$:*

*(i) (Marginal.) $\mathbb{P}(S_0 \le \hat{q}) \;\ge\; 1 - \alpha - d_{\mathrm{K}}(P, Q) - (n+1)\beta(b)$.*

*(ii) (Conditional.) For any $\delta, \delta' \in (0, 1)$, with probability at least $1 - \delta - n\beta(b)$ over $(\mathcal{D}_{\text{fit}}, S_{1:n})$,*

$$\mathbb{P}\big(S_0 \le \hat{q} \,\big|\, \mathcal{D}_{\text{fit}}, S_{1:n}\big) \;\ge\; b_{n,k}(\delta) - d_{\mathrm{K}}(P, Q) - \beta(b)/\delta',$$

*the last term holding on an event of probability at least $1 - \delta'$. Moreover $b_{n,k}(\delta) \ge 1 - \alpha - \sqrt{\log(1/\delta)/(2n)}$, and $b_{n,k}(\delta) \approx 1 - \alpha - \sqrt{2\alpha(1-\alpha)\log(1/\delta)/n}$, which is about 2.3 times tighter than the DKW form at $\alpha = 0.05$.*

*No assumption is made that the model underlying $\pi$ is correct.*

[Council correction: coupling cost is $(n+1)\beta(b)$ (Yu 1994, Lemma 4.1: $(m-1)\beta(b)$ for $m$ blocks, plus independence from $\mathcal{D}_{\text{fit}}$ and the deployment block); the conditional statement cannot be obtained from a joint total-variation bound and uses the $\beta$-mixing definition with Markov's inequality; the conformal coverage is $\mathrm{Beta}(k, n+1-k)$-distributed (Vovk 2012), so DKW is a corollary, not the tool; the drift penalty is Kolmogorov distance because only half-lines are evaluated, applied to the fixed set $(-\infty, \hat{q}^*]$ after coupling has made $S_0^*$ independent of $\hat{q}^*$.]

*Proof route.* (i) Sequential Berbee coupling replaces the $n+1$ separated blocks and $\mathcal{D}_{\mathrm{fit}}$ by independent copies $S^*_{0:n}$ on an event of probability $\geq 1-(n+1)\beta(b)$. (ii) On that event, conditional on $\pi$, the calibration copies are i.i.d. $P$ and $F_P(\hat{q}^*) \sim \mathrm{Beta}(k, n+1-k)$, so $\mathbb{P}(S^*_0 \leq \hat{q}^*) \geq 1-\alpha$ if $S^*_0 \sim P$. (iii) For $S^*_0 \sim Q$ and the fixed half-line $(-\infty, \hat{q}^*]$, $|Q(S \leq \hat{q}^*) - P(S \leq \hat{q}^*)| \leq d_{\mathrm{K}}(P,Q)$. Sum.

*Remark.* $\beta(b)$ is not estimable from data (Adams–Nobel-type impossibility). The certificate is conditional on an *assumed* mixing class $\{\beta(k) \leq Ck^{-r}\}$, and the thesis says so. [Council correction: added]

**Proposition 7** (Drift penalty bounded by $W_1$; target)**.** *If $Q$ has a density bounded by $L$ (no assumption on $P$), then for all $x$, $F_P(x) - F_Q(x) \leq \sqrt{2L\,W_1(P,Q)}$. If both have densities bounded by $L$, $d_{\mathrm{K}}(P,Q) \leq \sqrt{L\,W_1(P,Q)}$. Also $F_P(x) - F_Q(x) \leq L\,W_\infty(P,Q)$.*

*Proof.* Let $D = F_P - F_Q$. If $D(x_0) = d > 0$ then, since $F_P$ is non-decreasing and $F_Q$ is $L$-Lipschitz, $D(x) \geq d - L(x-x_0)$ on $[x_0, x_0 + d/L]$, whence $W_1(P,Q) = \int |D| \geq d^2/(2L)$. The $W_\infty$ bound uses the monotone coupling: $F_P(x) \leq F_Q(x + W_\infty) \leq F_Q(x) + LW_\infty$. ☐

[Council correction: the first version claimed $|F_P - F_Q| \leq LW_1$, which is false: $P = \delta_0$, $Q = U[0, 1/L]$ has $W_1 = 1/(2L)$ and CDF gap 1. With the square-root form the plug-in penalty decays at $n^{-1/4}$, not $n^{-1/2}$. The i.i.d. rate $\mathbb{E}W_1(P_n, P) \leq n^{-1/2}J_1(P)$ is Bobkov–Ledoux (2019) in $d=1$; deployment scores are dependent, so a $\beta$-mixing version (Dedecker–Merlevède type) is needed and is part of Theorem 9(v). $L$ is an assumption on the unobserved $Q$ and is stated as such.]

**Theorem 8** (Sharpness of the drift penalty)**.** *Let $T$ be any threshold procedure independent of the deployment score, and $C_T(R) := \mathbb{P}_{S\sim R}(S \leq T)$. For every $P$ with continuous distribution function and every $\varepsilon \in (0,1)$:*

*(a) (Universal.)* $\inf_{Q:\, d_{\mathrm{TV}}(P,Q) \leq \varepsilon} C_T(Q) \;\geq\; C_T(P) - \varepsilon.$

*(b) (Attained.) Let $Q_M$ be $P$ with its bottom-$\varepsilon$ mass on $(-\infty, q_\varepsilon(P)]$ moved to the point $M$. Then $d_{\mathrm{TV}}(P, Q_M) = d_{\mathrm{K}}(P, Q_M) = \varepsilon$ and*

$$C_T(Q_M) \;\leq\; C_T(P) - \varepsilon + \varepsilon\,\mathbb{P}\big(F_P(T) < \varepsilon\big) + \varepsilon\,\mathbb{P}(T \geq M).$$

*For split conformal, $\mathbb{P}(F_P(T) < \varepsilon) = \mathbb{P}(\mathrm{Beta}(k, n+1-k) < \varepsilon)$, exponentially small for $\varepsilon < 1-\alpha$. Hence the constant 1 on $d_{\mathrm{K}}$ in Theorem 6 is sharp.*

*Proof.* (a) is the definition of total variation applied to the event $\{S \leq T\}$ conditional on $T$. (b) Under $Q_M$, mass $\varepsilon$ that $P$ placed below every plausible threshold now sits at $M$; if $T < M$ and $F_P(T) \geq \varepsilon$, coverage falls by exactly $\varepsilon$. The two correction terms cover the complementary events. ☐

[Council correction: $\delta_{+\infty}$ is not a probability measure on $\mathbb{R}$. The far-mass construction $(1-\varepsilon)P + \varepsilon\delta_M$ gives only $(1-\varepsilon)C_T(P)$, which is weaker and sharp only for degenerate procedures. Moving mass from the bottom of $P$ to a far point is what attains the constant 1. Since $Q$ is never observed, this is a direct adversarial construction, not Le Cam's method; Le Cam enters only in Theorem 9(iv) where deployment scores are partially observed.]

**Theorem 9** (Calibration-conditional coverage error over a mixing class: optimal blocking and minimax rate; target, the doctoral theorem)**.** *Fix the mixing class $\mathcal{B}_r := \{\beta(k) \leq Ck^{-r}\}$, $r > 1$, and a calibration record of $N$ periods, stationary on the record with continuous marginal score law $P$; write $F_N$ for the empirical distribution function of the $N$ scores. A threshold procedure is any measurable $T = T(S_{1:N})$. Its conditional coverage error is $\mathcal{E}(T) := |F_P(T) - (1-\alpha)|$, the error in the coverage $F_P(T)$ received by a deployment score independent of the record; the* *deployment gap and its coupling cost are handled as in Theorem 6 and are excluded here, as is drift. The error is two-sided because the one-sided deficit is degenerate over all procedures ($T = +\infty$ has deficit zero); over-coverage is the efficiency loss. The coverage error at confidence $\delta$ is $\mathcal{E}_\delta(T) := \inf\{\varepsilon : \mathbb{P}(\mathcal{E}(T) > \varepsilon) \le \delta\}$ and the minimax value is $\mathcal{E}^*_\delta(N, r) := \inf_T \sup_{\mathcal{B}_r} \mathcal{E}_\delta(T)$.*

(i) *(Optimal block length for the blocked route; provable.) Blocks of length $b$ with gaps $b$ give $n = \lfloor N/(2b) \rfloor$ blocks. By Theorem 6(ii), the $\mathrm{Beta}(k, n+1-k)$ law of $F_P(\hat q)$ on the coupling event integrated over the confidence level, the expected conditional shortfall of $\hat q = S_{(k)}$ obeys*

$$\mathbb{E}\big[(1-\alpha-F_P(\hat q))^+\big] \;\le\; \frac{1}{2\sqrt{n+2}} + (n+1)\beta(b) \;\le\; \sqrt{\frac{b}{2N}} + C\,N\,b^{-(r+1)} \qquad (2 \le b \le N/2),$$

*so that, up to the constants $1/\sqrt{2}$ and $C$, the bound is $f(b) := \sqrt{b/N} + Nb^{-(r+1)}$. The exact minimiser of $f$ is*

$$b^* = (2(r+1))^{2/(2r+3)}\,N^{3/(2r+3)}, \qquad f(b^*) = \frac{2r+3}{2(r+1)}\,(2(r+1))^{1/(2r+3)}\,N^{-r/(2r+3)},$$

*with $f(b^*) = 1.51\,N^{-r/(2r+3)}$ and $b^* = 23.9$ at $r = 2$, $N = 500$; the bare balance $\sqrt{b/N} = Nb^{-(r+1)}$ gives $N^{3/(2r+3)} = 14.3$ and is not the minimiser.*

(ii) *(The blocked rate is not minimax; provable.) Since $|\operatorname{Cov}(\mathbf{1}\{S_0 \le x\}, \mathbf{1}\{S_k \le x\})| \le \alpha(k) \le \beta(k)$, $\operatorname{Var} F_N(x) \le (\frac{1}{4} + 2C\zeta(r))/N$ for every $x$, and the unblocked sample quantile $T_N := S_{(\lceil(1-\alpha)N\rceil)}$ of the whole record satisfies, for every $\varepsilon > 0$,*

$$\mathbb{P}\big(|F_P(T_N) - (1-\alpha)| > \varepsilon\big) \;\le\; \frac{\frac{1}{2} + 4C\zeta(r)}{N\varepsilon^2}, \qquad \text{hence} \qquad \mathcal{E}_\delta(T_N) \;\le\; \sqrt{\frac{\frac{1}{2} + 4C\zeta(r)}{N\delta}}.$$

*I.i.d. scores lie in every $\mathcal{B}_r$, and the two-point argument for them gives $\mathcal{E}^*_\delta(N, r) \ge c_\delta N^{-1/2}$. At fixed $\delta$ the minimax rate over $\mathcal{B}_r$ is therefore exactly $N^{-1/2}$, attained without blocking; the blocked rate $N^{-r/(2r+3)}$ of (i) is slower by the factor $\sqrt{b^*}$. Blocking buys the exact Beta law and a certificate whose only dependence on $(C, r)$ is the failure probability $(n+1)\beta(b)$; it does not buy rate.*

(iii) *(Mixing-driven lower bound at high confidence; target.) There are $c_r, \delta_0 > 0$ and, for every $N$, a stationary process in $\mathcal{B}_r$ (regime-renewal construction: i.i.d. regime labels with score laws $P_1 \ne P_2$, held for i.i.d. durations with tail $k^{-(r+1)}$, so that $\beta(k) \le \mathbb{P}(\text{no renewal in } k \text{ steps}) \asymp k^{-r}$) such that for every threshold procedure $T$ and every $\delta \in [N^{-r}, \delta_0]$,*

$$\mathbb{P}\big(\mathcal{E}(T) \ge c_r\,(N^r\delta)^{-1/(r+1)}\big) \;\ge\; \delta.$$

*Together with the Fuk–Nagaev inequality for bounded strongly mixing sequences (Rio 2017, Ch. 6) applied to $F_N$ at the two quantiles $q_{1-\alpha\mp\varepsilon}(P)$, this gives*

$$\mathcal{E}^*_\delta(N, r) \;\asymp\; \max\Big\{N^{-1/2} g_r(\delta),\; (N^r\delta)^{-1/(r+1)}\Big\}, \qquad \sqrt{\log(1/\delta)} \;\lesssim\; g_r(\delta) \;\lesssim\; \delta^{-1/r},$$

*and the mixing term binds exactly when $\delta < N^{-(r-1)/2}$ (at $N = 500$, $r = 2$: $\delta < 0.045$, so every certificate at the usual confidence levels sits in the mixing-bound regime). The same construction gives an expected shortfall of order $N^{-r/(r+1)}$, the rate that Barber and Pananjady (2026) prove for split conformal; part (iii) extends the lower bound to all threshold procedures, which they leave open.*

(iv) *(Partially observed drift; target.) When $m$ deployment scores are additionally observed, the minimax certifiable deficit over $\{Q : d_{\mathrm{K}}(P, Q) \le \varepsilon\}$ is obtained adaptively in unknown $\varepsilon$ by Le Cam's two-point method on the observed deployment sample.*

(v) *(Dependent empirical $W_1$; cited and specialised.) The dependent analogue of the Bobkov–Ledoux bound, $\mathbb{E}\, W_1(Q_m, Q) \lesssim m^{-1/2}$ under summable dependence with a moment condition on $Q$, is Dedecker and Merlevède (2017). The thesis specialises it to $\beta \in \mathcal{B}_r$, $r > 1$, with an explicit $c_r$, closing the gap in Proposition 7. This part is not claimed as new.*

[Council correction: third version. The second version stated (ii) as "no threshold procedure achieves deficit smaller than $cN^{-r/(2r+3)}$; hence the rate is minimax over $\mathcal{B}_r$". That is false in every coverage formulation: marginally, Barber and Pananjady (2026, Cor. 1–2) reach $N^{-r/(r+1)}$ without blocking; conditionally, the unblocked sample quantile reaches $N^{-1/2}$ for all $r > 1$ by the covariance inequality, and over all procedures the one-sided deficit is degenerate. The second version also derived $\sqrt{b/N} + Nb^{-(r+1)}$ "by Theorem 6(i)", which has no $\sqrt{b/N}$ term; the concentration term exists only in the conditional form (ii). The bare balance was quoted as $b^*$; the exact minimiser carries $(2(r+1))^{2/(2r+3)}$ (code agent, 2026-09-09). The old part (iii) is Dedecker–Merlevède (2017) and is downgraded to cited.]

*Remark* (Decision: why the spine is the calibration-conditional error)*.* Three options were weighed. (a) Keep marginal coverage and target the Barber–Pananjady open question at rate $N^{-r/(r+1)}$: rejected as the spine, because over all procedures the marginal deficit is degenerate without an efficiency constraint, and adding one turns the question into (b); the blocked marginal bound of Theorem 6(i) is dominated by their unblocked bound and is retained only as the route to (ii). (b) Restate for the calibration-conditional coverage error at confidence $\delta$: chosen. The concentration term is real there, the two-sided error is non-degenerate over all procedures, the fixed-$\delta$ minimax rate is settled by part (ii), and the high-confidence regime of part (iii) is where the mixing exponent is provably the binding constraint, which is the object a model-risk certificate needs. (c) Shift the spine to dependent $W_1$ plus partially observed drift: rejected, because Dedecker and Merlevède (2017) already give the dependent $W_1$ rate and the drift part is an adaptation of Le Cam's method. Positioning: Halkiewicz (2026) proves an optimal calibration window with a Le Cam lower bound for rolling-origin conformal under local stationarity and $\alpha$-mixing; his hardness is drift-driven and his class carries no $\beta$-mixing exponent, no blocking and no conditional coverage. Theorem 9 is not the first optimal-design result for conformal prediction under dependence and must not be described as such; its claim is the mixing-driven minimax statement (iii) over $\mathcal{B}_r$ for conditional coverage, and the honest accounting (ii) of what blocking does and does not buy.

*Proof route.* (i) $\mathbb{E}[(1-\alpha-B)^+] \le \mathbb{E}|B - k/(n+1)| \le \mathrm{sd}(B) \le 1/(2\sqrt{n+2})$ for $B \sim \mathrm{Beta}(k, n+1-k)$, since $k/(n+1) \ge 1-\alpha$; add the coupling failure $(n+1)\beta(b)$ of Theorem 6; differentiate $f$. (ii) $F_P(T_N) < 1-\alpha-\varepsilon$ forces $F_N(q_-^-) - F_P(q_-^-) \ge \varepsilon$ at $q_- = q_{1-\alpha-\varepsilon}(P)$, and $F_P(T_N) > 1-\alpha+\varepsilon$ forces $F_N(q_+) - F_P(q_+) < -\varepsilon$; Chebyshev with the covariance bound at each point; the i.i.d. lower bound is the two-point argument on Bernoulli data with $\mathrm{KL} \asymp N\varepsilon^2$. (iii) A holding interval of length $m \asymp (N/\delta)^{1/(r+1)}$ meets the record with probability $\asymp Nm^{-(r+1)} \asymp \delta$ and shifts $F_N$ at the quantile by $\asymp (m/N)\, d_{\mathrm{K}}(P_1, P_2)$; to pass from the sample quantile to all procedures, compare two regime-renewal processes whose mixture weights differ by $\asymp m/N$ and whose $N$-sample laws are close in total variation; Fuk–Nagaev (Rio 2017, Thm 6.2 form: a Gaussian-type term $(1 + N\varepsilon^2/(r\sigma^2))^{-r/2}$ plus a polynomial term $N^{-r}\varepsilon^{-(r+1)}$) for the upper bound. (iv) Two-point reduction on the deployment sample. (v) Specialise Dedecker–Merlevède to blocked scores under $\mathcal{B}_r$ with the Berbee coupling of Theorem 6.

*Failure mode.* (i) and (ii) are elementary and cannot fail. In (iii) the two-point reduction over all procedures may lose a logarithmic factor or hold only over the regime-renewal subclass of $\mathcal{B}_r$; the $\delta$-dependence $g_r$ between $\sqrt{\log(1/\delta)}$ and $\delta^{-1/r}$ is open and is stated as open. Either outcome is publishable; the fixed-$\delta$ rate is settled either way.

*Remark* (On the first-draft bootstrap theorem)*.* The comparison with the stationary block bootstrap is no longer stated as a theorem. Bootstrap coverage rates for quantiles under mixing require Edgeworth expansions with unknown constants, and "the only method with a provable finite-sample level" is a comparison of assumptions, not a result. The comparison is retained as an empirical section on USD and ZAR data. [Council correction: dropped]

## 4 Difficulty after correction

| Result | Kind | Deepest tool | Difficulty (1–5) | Carries the doctorate |
|---|---|---|---|---|
| Lemma 1 | cited | deterministic time change | 1 | no |
| Prop. 2 | derivation | HLW effective parameters | 2 | no |
| Thm 3 | expansion, error bound | forward measure, SABR inversion | 3 | finance paper |
| Thm 4 | corollary | nonsingular Jacobian | 2 | no |
| Prop. 5 | algebra | variance matching | 2–3 | appendix |
| Thm 6 | theorem | Berbee coupling, Beta law, $d_{\mathrm{K}}$ | 3 | supports |
| Prop. 7 | proposition | Lipschitz CDF argument | 2 | supports |
| Thm 8 | theorem | adversarial construction | 2 | supports |
| **Thm 9** | **theorem** | **regime-renewal lower bound, Fuk–Nagaev under mixing** | **4** | **yes: the spine** |

Not needed anywhere: Malliavin calculus, rough paths, G-expectation, optimal transport, reflected BSDEs, deep-learning theory.

## 5 The three-year programme

**Principles.** The statistics spine is proved first, in the candidate's home field, while finance prerequisites are learned in parallel: Theorem 6 needs only a fixed policy, and the market convention (strike shift, volatility scaling) supplies one without Theorem 3. The USD LIBOR-to-SOFR transition is the empirical spine, because the target smile became observable there and every input is on the university terminal; the South African book is a coda, because by month 30 it will have at most twenty-four monthly hedging-error observations and its option surfaces need a sponsor who may not sign.

**Cuts.** Proposition 5 to an appendix, stated with the exact case (a) proved and (b) as a conjecture. The first-draft bootstrap theorem to an empirical section. The standalone factual-record note folded into Chapter 1 at three thousand words. The standalone time-changed-SABR note becomes Chapter 2 preliminaries. The South African chapter shrinks to five thousand words.

**Minimal passing thesis.** Theorems 6, 8 and 9 proved; the USD certificate test with honest coverage; Theorem 3 with the $O(\varepsilon)$ identified part and Theorem 4 demonstrated numerically on USD; the South African coda on public data if no sponsor. This passes because the spine is proved, sharp, rate-optimal, and tested where the truth was observable.

| Months | Activity | Milestone |
|---|---|---|
| 1–4 | Stochastic calculus and forward measures; Barber et al. (2023), Oliveira et al. (2024), Barber and Pananjady (2026), Vovk (2012), Rio (2017, Ch. 6) line by line | Reading report |
| 3–6 | Theorem 8 proved; Theorem 6 route drafted with the Beta form | Thm 8 done (m5) |
| 5–9 | USD multi-curve bootstrap and time-changed SABR caplet code; Lemma 1, Prop. 2 | Working code |
| 7–10 | Theorem 6 full proof; Proposition 7 | Thm 6 done (m10) |
| 9–11 | Proposal defence: Chapters 1–2 drafts, Theorems 6, 8 proved, Theorem 9 stated | Faculty approval (m11) |
| 10–12 | USD data assembled; ZAR sponsor decision | Data agreement or public-data fallback (m12) |
| 10–18 | Theorem 9: (i)–(ii) proved (m12), regime-renewal lower bound (iii) (m16), Fuk–Nagaev upper bound, (iv)–(v) (m18) | Thm 9 done (m18) |
| 12–17 | USD certificate test with optimal blocking; bootstrap and traffic-light comparison | Chapter 6 part one |
| 14–20 | Theorem 3 expansion; Theorem 4 construction, with finance co-supervisor | Chapter 4 draft (m20) |
| 18–22 | Paper 1: Theorems 6, 8, 9, Prop. 7, USD test | Submitted, *EJS* or *Bernoulli* (m22) |
| 20–24 | USD identifiability validation: predicted SOFR smile and $\pm\eta$ interval against the observed smile | Chapter 6 complete (m24) |
| 24–28 | Paper 2: Theorems 3, 4, USD identifiability | Submitted, *Quantitative Finance* (m28) |
| 26–30 | ZAR coda: JIBAR surface to Dec 2026, converted-book P&L to m30, reserve decomposition | Chapter 7 draft (m30) |
| 30–34 | Full write-up; Chapters 1, 8, appendices; internal review | Complete draft (m34) |
| 34–36 | Corrections, submission | Thesis submitted (m36) |

**Chapters and words.** 1 Introduction and the transition (5k); 2 Preliminaries (8k); 3 Certification (10k); 4 Sharpness and optimal blocking (6k); 5 Identifiability (10k); 6 USD validation (10k); 7 South African coda (5k); 8 Conclusions (3k); appendices. About 57 thousand words.

**Papers.** Two, not three. Paper 1 is the doctoral paper. The South African material becomes a practitioner note to the Market Practitioners Group.

**Probability of submission.** Examiner's estimate: about 60% by month 36; about 85% by month 42 with the customary extension. The plan passes without ZAR data and without Theorem 3 being clean. It does not pass without Theorem 6 by month 10–12, Theorem 9 by month 18, and the USD test by month 17.

**Ranking under three years.** Three years widens this topic's lead over the alternatives the council reviewed. Conformal deep hedging on index options needs a year of training infrastructure before any theorem. Certified superhedging under volatility uncertainty needs G-expectation prerequisites and carries difficulty-5 theorems that can fail outright. This topic's spine uses tools half-known already, and its finance half, which is the schedule risk, is exactly what the re-cut trims.

# References


- Barber, R. F., Pananjady, A. (2026). Predictive inference for time series: why is split conformal effective despite temporal dependence? *Proc. 37th Int. Conf. Algorithmic Learning Theory*, PMLR 313, 1–24. arXiv:2510.02471.
- Barber, R. F., Candès, E. J., Ramdas, A., Tibshirani, R. J. (2023). Conformal prediction beyond exchangeability. *Ann. Statist.* 51(2), 816–845.
- Bobkov, S., Ledoux, M. (2019). One-dimensional empirical measures, order statistics, and Kantorovich transport distances. *Mem. Amer. Math. Soc.* 261.
- Dedecker, J., Merlevède, F. (2017). Behavior of the Wasserstein distance between the empirical and the marginal distributions of stationary $\alpha$-dependent sequences. *Bernoulli* 23(3), 2083–2127.
- Hagan, P. S., Kumar, D., Lesniewski, A. S., Woodward, D. E. (2002). Managing smile risk. *Wilmott*, Sept., 84–108.
- Halkiewicz, S. M. S. (2026). Rolling-origin conformal prediction under local stationarity and weak dependence. arXiv:2605.08422.
- Lyashenko, A., Mercurio, F. (2019). Looking forward to backward-looking rates. SSRN 3330240.
- Oliveira, R. I., Orenstein, P., Ramos, T., Romano, J. V. (2024). Split conformal prediction and non-exchangeable data. *J. Mach. Learn. Res.* 25.
- Piterbarg, V. (2020). Interest rates benchmark reform and options markets. SSRN 3537925.
- Rio, E. (2017). *Asymptotic Theory of Weakly Dependent Random Processes.* Probability Theory and Stochastic Modelling 80, Springer.
- Vovk, V. (2012). Conditional validity of inductive conformal predictors. *Proc. ACML*, JMLR W&CP 25, 475–490.
- Willems, S. (2020). SABR smiles for RFR caplets. arXiv:2004.04501.
- Yu, B. (1994). Rates of convergence for empirical processes of stationary mixing sequences. *Ann. Probab.* 22(1), 94–116.